

v1.0Kaon Deep Dive_ A Computational Model

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Kaon Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026 **Applies to:** v1.0Kaon Deep Dive_ A Computational Model V2.0.md

Revision Necessity — Four Drivers

- Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
- New confirmed particles and phenomena** — Pines’ Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
- Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^{7+} particles
- Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
NA62 rare kaon BR	$K \rightarrow \pi \nu \bar{\nu}$ predicted BR = 8.6×10^{-11}	NA62 2024: measured BR = $12.6 \pm 2.7 \times 10^{-11}$ — 2.3σ above SM. BSM tension flag added	HIGH	Possible BSM signal requires tracking in simulation
Epsilon_K CP violation	PDG 2020 era	RBC+UKQCD 2022: SM $\epsilon_K = 1.81 \times 10^{-3}$, exp = 2.228×10^{-3} . 2.5σ tension — update kaon mixing	MEDIUM	CP violation magnitude tension requires parameter update
CKM matrix 2024	V1.0 era CKM values	CKMfitter 2024: $\alpha = 85.4^\circ$, $\beta = 21.9^\circ$, $\gamma = 72.7^\circ$. More precise triangle closure	LOW	Precise CKM elements reduce kaon oscillation parameter uncertainty
Oscillation visualization	Color oscillation concept	V2.0: K_0 instances oscillate $K_1 \leftrightarrow K_2$ at exact $\Delta m_K = 3.484 \times 10^{-12}$ MeV period (1.19 ns). Green/red color cycle	MEDIUM	Physical oscillation period now precisely implemented in rendering

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** NA62
 - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
 - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
 - **New interaction modes:** Triggered by new experimental discoveries
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Unchanged Architecture

All foundational philosophy is **carried forward unchanged:** - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric ($C \cdot F \cdot S$ product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Kaon Deep Dive_ A Computational Model V1.0 archived | V2.0 is the active specification as of June 2026